

TRƯỜNG ĐẠI HỌC FPT

CAPSTONE PROJECT REPORT

Report 5 – Software Testing Documentation

V. Software Testing Documentation

1. Scope of Testing

Testing covers the SME-FraudGuard workflows, role boundaries and quality attributes defined in the registration proposal and the approved SRS. The scope includes Platform Admin, Implementation Staff, SME Admin, Operator, Investigator, customer transaction systems and external AI, payment and notification integrations.

Functional coverage includes authentication and password changes; customer/workspace administration; Staff assignment; onboarding documents, mapping, rule setup and sample tests; readiness, approval and activation; API-key issue and disablement; REST API, incoming Webhook and CSV/Excel ingestion; validation, deduplication and import results; plan-based detection and fallback; alert review, case assignment, investigation reports and Operator conclusions; dashboards, exports, notifications, audit history; and plan purchase, renewal and payment confirmation.

Cross-cutting tests verify role-, workspace- and object-based authorization, sensitive-data protection, separation of sample and production processing, preservation of accepted data and safe repeated requests. Quality testing follows SRS 4.2: Security and Authorization, Tenant Isolation and Reliability, and Usability.

The scope excludes automatic customer-account blocking or transaction cancellation, mandatory AI confidence/explanations/risk scores, automatic model retraining/deployment and commercial transaction quotas. Model-quality evaluation requires a separately approved dataset and metrics. Performance/load thresholds and recovery-time targets are not introduced without confirmed requirements.

2. Test Strategy

Tests are derived from use cases, business rules, common requirements and the three SRS quality attributes. Risk-based prioritization places authorization, tenant isolation, transaction integrity and payment entitlement first. Core workflow checks follow, then lower-impact presentation checks.

Each scenario records prerequisites, controlled data, numbered actions and observable expected outcomes. Positive, negative, boundary, retry, concurrent and dependency-failure variants are included where the confirmed behavior requires them. Unresolved requirements remain explicit decisions; independent scenarios proceed without selecting an assumed answer.

2.1 Testing Types

Functional testing: verify permitted actions, validation, lifecycle prerequisites, user-visible messages and persisted business outcomes across the complete workflow.

Security and authorization testing: attempt cross-role, cross-workspace and unassigned-object access through APIs, UI, exports, API keys and integration callbacks. Verify safe responses, one-time key disclosure, default masking and authorized audited plaintext access.

Reliability and data-integrity testing: exercise duplicate/conflicting transactions across channels, repeated payment confirmations, concurrent acceptance, failed processing after acceptance, AI fallback and temporary dependency failures. Confirm preserved records and absence of repeated initial effects.

Usability testing: inspect role-specific navigation, actionable validation errors, import row summaries and clear distinctions among AI/Rule Alerts, detection outcomes and human conclusions. Verify safe display and export of user-provided text.

Regression testing: rerun the affected workflow and its shared authorization, ingestion or entitlement checks after a change or defect fix. Use configured technical boundaries; do not invent numeric nonfunctional targets.

2.2 Test Levels

Unit testing checks isolated domain decisions and transformations, including mapping/conversion, typed rule operators, optional-value handling, plan-date calculations and authorization predicates. Detailed unit cases are linked to the finalized implementation contracts.

Integration testing checks backend persistence and module boundaries, ingestion adapters, AI-result validation, rule fallback, payment verification and notification delivery using controlled dependencies. Database-backed tests check tenant filtering, atomic uniqueness and preservation of accepted data.

System testing exercises complete role-based journeys through the deployed test system and public integration boundaries. It includes misuse, expired rights, duplicate/conflict, partial-import and failure-recovery scenarios.

Acceptance testing reviews the agreed business journeys and proposal/SRS criteria with the project team and supervisor. Acceptance uses recorded evidence from a known build and environment; a designed test case alone is not acceptance evidence.

2.3 Supporting Tools

JUnit 5 and Maven support Java unit and integration tests. ArchUnit checks the modular-monolith dependency rules; Testcontainers provides disposable PostgreSQL environments for database-backed checks.

An API client or scripted HTTP runner exercises protected endpoints, retries, integration verification and negative requests. Browser tools support end-to-end workflow and usability checks; UI state is cross-checked against authorized API or persisted outcomes.

Controlled AI responses, a payment-gateway sandbox or verification test double, and a notification sink provide repeatable external-service variants. Failure injection is restricted to the isolated test environment.

The Excel system-test workbook maintains procedures, traceability, actual results and evidence references. Version-controlled synthetic CSV/XLSX/JSON fixtures provide repeatable business inputs. Screenshots, sanitized responses and logs identify the build, scenario and observed result without credentials, keys or sensitive plaintext.

3. Test Plan

3.1 Human Resources

3.2 Test Environment

3.3 Test Milestones

4. Test Cases

4.1 Unit Test Cases

4.2 System Test Cases

The system-test catalogue is maintained in the linked workbook. It contains 170 designed cases across 17 groups, with references to 42 use cases, 40 business rules, 19 common requirements and SRS 4.2 quality attributes. These counts express designed references, not executed or passed requirements.

System Test Cases Workbook

Synthetic Test Data Package

Each case includes its priority, preconditions, test-data references, procedure and expected results. Actual result, evidence, tester, run date and defect reference are completed during execution. Status is Not Run until executed; Pass, Fail, Blocked and Not Applicable require a recorded outcome or reason.

Transaction fields, types and date/time conversion follow each workspace's configured mapping; no universal field list or date format is prescribed here. TEST-MAP-A is explicitly synthetic test configuration. The mixed file requires TX-BASE to be accepted first and expects 2 accepted, 1 duplicate, 1 conflict and 3 invalid rows. Independent cases use fresh transaction references.

Recovery checks cover fallback events from a historical Subscription period, reanalysed before or after current expiry. The Decisions sheet retains unanswered alert reconciliation and scheduling questions; the global input contract stays deferred and reporting remains generic. Detailed FR traceability follows the finalized SRS section 3. The following overview groups the designed scenarios.

Test group	Cases	Main checks
Authentication	14	Role-specific sign-in, initial password change, invalid credentials, session access and logout.
Access boundaries	12	Tenant/object tampering, Staff reassignment, Investigator assignment and role revocation.
Platform administration	9	Workspace/customer setup, assignment, members, permitted metadata and access restrictions.
Onboarding	14	Documents, sample isolation, readiness, approval, activation and initial plan timing.
API keys	10	Authorized issue/replace/disable, one-time disclosure, submission scope and revoked-key rejection.
Configuration	3	Workspace mapping, role restrictions and one active configuration set.
Rules	14	Typed operators, AND/OR, empty optional values, active rules and forbidden rule capabilities.
Transaction ingestion	17	REST/Webhook inputs, mapping, validation, same-ID retries/conflicts and concurrent acceptance.
Production import	10	CSV/XLSX authorization, structural errors, mixed rows, summaries and recovery after acceptance.

Test group	Cases	Main checks
Detection	12	Free rules, Subscription AI, Normal/Anomaly, response validation, fallback and preserved history.
Cases and investigation	13	AI-only case eligibility, subscription restrictions, assignment, reports and Operator conclusions.
Reporting	7	Scoped dashboards, history, filters, exports and safe failure handling.
Privacy	7	Masking, object/field-scoped disclosure and audit without sensitive plaintext.
Notifications	2	Correct recipient/workspace and isolated delivery failures.
Platform monitoring	1	Operational health visibility without enterprise production content.
Common behavior	5	Traceable audit, concurrent changes, time display, safe rendering and error handling.
Subscription and billing	20	Catalogue, SME Admin purchase/renewal, verified payments, retries, calendar dates and expiry.

5. Test Reports

5.1 Unit Test Report

5.2 System Test Report